Market Concentration Methods Note

HHI, CR4, and scenario analysis

Calculation Steps

- Convert market shares to decimals and verify that firm shares are internally consistent.
- Aggregate brands or products to the firm level when HHI is intended to measure firm concentration.
- Square each firm share.
- Sum squared shares and multiply by 10,000 to express HHI on the common 0-10,000 scale.
- Calculate CR4 by summing the four largest firm shares.
- For a merger scenario, combine the shares of the merging firms and recompute the HHI.

Browser Case

The browser case is calculated at the firm level. This matters because Microsoft had multiple browser products in the period. Treating products as separate firms would understate the firm-level concentration measure.

Metric	2012	2017	2022
HHI	2,772	4,398	4,759
CR4	96.6%	95.5%	95.0%
Largest firm share	32.8%	63.2%	66.9%

Issuer Case

The issuer case calculates a baseline HHI for named issuers and then models a scenario in which Capital One and Discover are combined. The resulting delta HHI equals the post-scenario HHI minus the base HHI.

Metric	Value
Base HHI - named issuers	886
Post-scenario HHI - named issuers	1,053
Delta HHI	+168
Base CR4	51.0%
Base CR5	60.9%

Limitations

Concentration metrics are useful screening tools, but they do not by themselves prove market power or predict competitive effects. A stronger analysis would also examine market definition, product substitutability, entry conditions, switching costs, pricing behavior, innovation, and consumer outcomes.